

C-STGB: Conformal Spatio-Temporal GraphBoost End-to-End Surveillance Architecture

STAGE 1: STREAMING INGEST

DuckDB Invariants & Hawkes

Continuous Ingest & Subgraphs

$$\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t), \mathcal{N}_K(u) (K \leq 15)$$

DuckDB / Arrow zero-copy micro-batching

Mass-Balance Flow Conservation

$$\tilde{\Phi}_{\text{flow}} = \log(1 + \frac{\sum_{\text{out}}^A}{\sum_{\text{in}}^A + \epsilon}) \approx 1.0$$

Detects pass-through mule dissipation

Hawkes Intensity & Causal Taint

$$\lambda_u(t) = \mu_u + \sum_{t_i < t} \alpha e^{-\beta \Delta t_i}$$

$\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$ localized taint (< 0.08 ms)

Out: $\mathbf{z}_{\text{inv}}(u, t) \in \mathbb{R}^{12}, \mathcal{G}_t^{(K)}$

Latency: 0.45 ms | DuckDB

STAGE 2: GNN & FILTER

Tri-Band & Latent GraphSMOTE

Tri-Band Continuous Attention

$$\Phi_{\text{time}}(\Delta t) \in \mathbb{R}^{d_t}, w(\Delta t) = \sum \pi_b e^{-\lambda_b \Delta t}$$

Resolves micro-bursts & 90-day dormancy

Context-Aware Edge Gating

$$\hat{g}_{ij} = \delta_{\text{floor}} + (1 - \delta_{\text{floor}}) \mathbf{g}_{ij}$$

Prunes 65.9% camouflage links

Typology Latent GraphSMOTE

$$\mathbf{h}_{\text{syn}} = (1 - \rho) \mathbf{h}_u + \rho \mathbf{h}_v, u, v \in C_k$$

Bilinear links $\hat{\mathbf{A}}_{\text{syn}} > \tau$; Hard-negatives

Out: $\mathbf{h}_u^{(L)} \in \mathbb{R}^d$ (Latent Manifold)

Continuous-Time Attention

STAGE 3: FUSION & RISK

Evidence-Adaptive Bayes Head

Evidence-Adaptive Fusion Gate

$$\alpha_u = \sigma(\mathbf{w}_\alpha^\top [\mathbf{h}_u^{(L)} \parallel \mathbf{z}_{\text{inv}} \parallel \mathbf{x}_u] + b_\alpha)$$

Cold-start deg(u) = 0 $\Rightarrow$ $\alpha_u \rightarrow 0$

Unified Representation Space

$$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u^{(L)} + (1 - \alpha_u) [\mathbf{z}_{\text{inv}} \parallel \mathbf{x}_u]$$

Maintains 89.6% F1 on isolated entities

Cost-Sensitive Bayes Risk Head

$$\tau^* = \arg \min_{\tau} (15 \cdot \text{FN} + 1 \cdot \text{FP})$$

Asymmetric Focal Tversky Loss $\mathcal{L}_{\text{task}}$

Out: $\hat{p}_u \in [0, 1], \mathbf{h}_u^* \in \mathbb{R}^{d^*}$

Cold-Start + Bayes Risk

STAGE 4: CONFORMAL SWARM

CRC Triage & FinCEN SAR

Class-Conditional CRC Quantiles

$$\hat{q}^{(y)} = \text{Quantile}(\mathcal{D}_{\text{cal}}), 1 - \alpha \geq 99.0\%$$

Finite-sample coverage; Online ACI drift

3-Tier Deterministic Triage

$$\Gamma(u) \in \{\{1\}, \{0, 1\}, \{0\}\}$$

Automates >99.4% streaming volume

Multi-Agent Forensic Swarm

Investigator $\rightarrow$ Auditor $\rightarrow$ Drafter

FinCEN Form 111 XML | Merkle SHA-256

Out: $\Gamma(u) \subseteq \{0, 1\}, \mathbf{XML}_{\text{SAR}}$

SR 26-2 / EU AI Act Certified